THE JOURNAL OF NAVIGATION (2018), **71**, 100–116. © The Royal Institute of Navigation 2017
doi:10.1017/S0373463317000546

Research on Ship Classification Based on Trajectory Features

Kai Sheng, Zhong Liu, Dechao Zhou, Ailin He and Chengxu Feng

(*College of Electronic Engineering, Naval University of Engineering, Wuhan, Hubei, China*)
(E-mail: shengkai0214@foxmail.com)

It is important for maritime authorities to effectively classify and identify unknown types of ships in historical trajectory data. This paper uses a logistic regression model to construct a ship classifier by utilising the features extracted from ship trajectories. First of all, three basic movement patterns are proposed according to ship sailing characteristics, with related sub-trajectory partitioning algorithms. Subsequently, three categories of trajectory features with their extraction methods are presented. Finally, a case study on building a model for classifying fishing boats and cargo ships based on real Automatic Identification System (AIS) data is given. Experimental results indicate that the proposed classification method can meet the needs of recognising uncertain types of targets in historical trajectory data, laying a foundation for further research on camouflaged ship identification, behaviour pattern mining, outlier behaviour detection and other applications.

KEYWORDS

1. Ship classification. 2. Trajectory feature extraction. 3. AIS data. 4. Logistic regression.

Submitted: 18 February 2017. Accepted: 10 July 2017. First published online: 23 August 2017.

1. INTRODUCTION. Automatic Identification System (AIS) base stations and coastal radar stations have been established in many coastal regions, generating a large amount of ship information every day. This includes not only static information such as ship type, tonnage, country, etc., but also dynamic information, such as time, location, direction, speed, and so on. Under normal circumstances, the real-time status of maritime vessels can be identified easily based on the above information, but marine traffic and ship behaviour patterns can also be deduced from historical data, so as to facilitate maritime supervision, abnormal behaviour detection and port infrastructure construction. However, some vessels with AIS cheat detection by turning off transponders, falsifying position data, or deliberately transmit improper identification data to evade detection for illegal operations or even illegal exploration (McCauley et al., 2016). In addition, there are still such problems as data quality and radar target recognition. For these reasons, it is often very difficult to find out the true type or identity of some ships in the historical database, which not only poses

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

a challenge in marine surveillance and maritime security defence, but also interferes with further data mining. Therefore, it has become an important task to classify and identify these unknown vessels correctly.

In recent years, the study of the classification and prediction of moving objects by using data mining technology and massive trajectory data (Zheng, 2015; Feng and Zhu, 2016) has become common. For instance, with regard to the classification of ground moving objects, such as pedestrians, bicycles and cars, Dodge et al. (2009) extracted global and local movement features from trajectories and took the Support Vector Machine (SVM) approach to train the classification model, which was used to predict the object type of unknown trajectories. Considering the classification and prediction of hurricane trajectories, Patel et al. (2012) and Patel (2013) proposed not only to take the spatial distributions and shapes of trajectories as features, but also to extract the duration-aware features from trajectories for building a classifier, achieving higher accuracy of classification than before.

Other researchers classified and predicted the activity patterns of moving objects by combining trajectory features with geographic and semantic information. For example, Zhu et al. (2011) aimed to infer the status of a taxi, consisting of *Occupied, Non-occupied*, and *Parked*, according to its trajectories, so they partitioned trajectories using a stay-point detection method, and then extracted from each segment a set of features based on historical trajectories, road networks and Points Of Interest (POI). After that, they used the identified features to train a local probabilistic classifier and then globally considered travel patterns via a hidden semi-Markov model. Boukhechba et al. (2015) presented an approach capable of incrementally extracting semantic locations from the trajectories of pedestrians, and achieved real-time classification and prediction of users' activity types based on an online K-means algorithm.

Currently, most research on ship trajectory data focuses on marine traffic pattern mining, ship collision risk assessment, and maritime anomaly detection (Aarsæther and Moan, 2009; Silveira et al., 2013; Zhen et al., 2017), but few researchers pay attention to ship classification. Lee et al. (2008) proposed a trajectory classification method using hierarchical region-based and trajectory-based clustering after trajectory partitioning. Regardless of the movement patterns, they first utilised region-based clustering to discover regions in which trajectories mostly belong to one class. Subsequently, trajectory-based clustering (Lee et al., 2007) was employed to discover sub-trajectories indicating common movement patterns of each class. By using the collaboration between two types of clustering, they could discover two types of discriminative features for identifying the type of the moving objects. Nevertheless, this method did not consider the essential differences between different types of moving objects. During the classification of ship types, this method worked efficiently only in regions where types of ships clearly differed, such as the vicinity of harbours or in fisheries. De Vries and Van Someren (2012; 2014) first compressed trajectories through two-stage piecewise linear segmentation to retain stop and move information. Then, they defined trajectory similarities based on compressed trajectories as kernels, and used them in a kernel method (such as SVM) for ship classification. If a ship arrives in a new area or where such types of ships have rarely appeared before, it is much more likely to be wrongly classified. However, this classification method is not suitable for ships sailing on a wide ranged sea area with a high degree of freedom. Elwakdy et al. (2015) extracted the polynomial coefficients of sub-trajectories as trajectory features after trajectory segmentation and polynomial fitting, and built a ship classifier based on an Adaptive Neuropathy-Fuzzy Inference System (ANFIS). The advantage of this method is to utilise

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

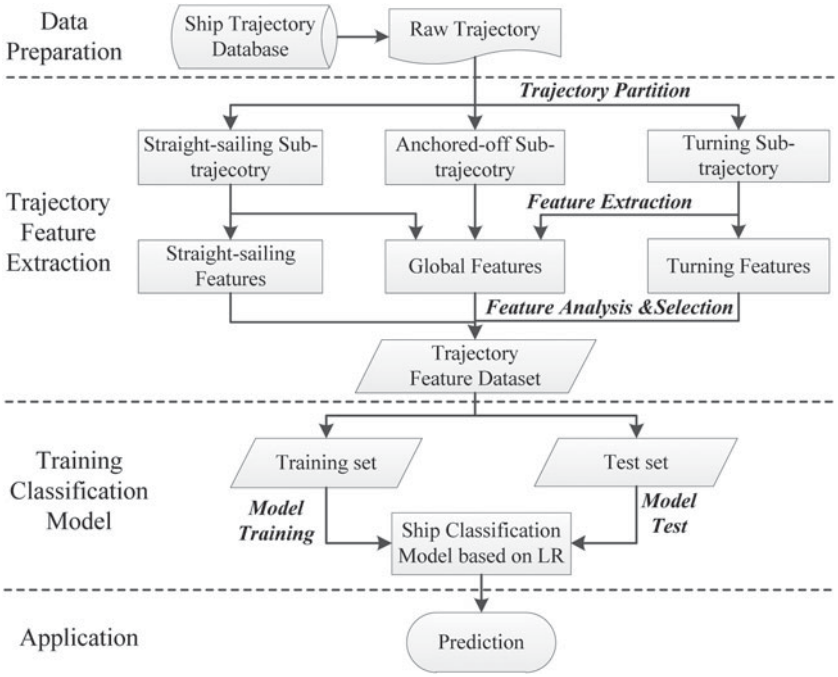

Figure 1. Overall process of building a ship classification model.

the innate shape characteristics of trajectories to avoid the influence of the geographical spaces of the ships. However, the temporal dimension is discarded, and it is required that segments of the same amount are divided from different types and lengths of ship trajectories. It is therefore difficult to ensure all trajectories can be partitioned properly when the data set is large or the duration is long. Failure to do so may lead to a poor fitting result with complex trajectories and loss of many effective features for classification.

To identify the types of ships more accurately in a broader sea area, this paper utilises the Logistic Regression (LR) model, one of the most popular classification techniques in the machine learning domain, to classify the ships. The overall process of building the classification model is shown in Figure 1. First, the raw trajectory is split into anchored-off, straight-sailing, and turning sub-trajectories based on the proposed three basic movement patterns and the related sub-trajectory partitioning algorithms. Second, three categories of trajectory features are extracted from the sub-trajectories on the basis of the presented feature extraction methods. Third, the trajectory feature dataset is generated by analysing and selecting the extracted features. Last, the ship classifier is constructed by using the LR model. In this paper, a case study is shown on the building of a classification model for identifying fishing boats and cargo ships from real AIS data. Experimental results indicate that the proposed methods can well meet the needs of recognising unclear types of targets from historical trajectory data.

The rest of this paper is organised as follows. Section 2 briefly introduces the logistic regression classification algorithm. Section 3 proposes ship movement patterns with the related trajectory partitioning algorithm, and the trajectory feature extraction method.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Section 4 presents an experiment on the classification of fishing boats and cargo ships based on real AIS data. Section 5 concludes the paper, and outlines potential future research.

2. LOGISTIC REGRESSION ALGORITHM.

The problem of ship type recognition based on trajectory features can be attributed to the classification problem in the machine learning domain. There are many classification algorithms, such as the Decision Trees (DT), Artificial Neutral Networks (ANNs), Logistic Regression (LR) and Support Vector Machines (SVMs). As shown in Table 1, each algorithm has its own advantages and disadvantages and suitable scope of application (Kotsiantis, 2007; Hastie et al., 2009; Harrington, 2012). Thanks to its advantages including high accuracy, simple mathematical model, easy programming and good interpretability, the logistic regression algorithm is chosen for our classification model. This section briefly introduces the logistic regression algorithm (Schein and Ungar, 2007; Zhou, 2016).

2.1. *Logistic regression classifier and sigmoid function.*

Logistic regression is a generalised linear model and is mainly used in two classification problems. We assume that the training dataset $\{(x_{i1}, x_{i2}, x_{i3}, ..., x_{iN}, y_i)\}_{i=1}^{M}$ is given, $X = (x_1, x_2, x_3, ..., x_n) \in R^N$ describes the N-dimensional feature vector and y is the classification label, whose value range is $\{0, 1\}$. We expect to build a linear regression model $z = w^T X + b$ for the classification boundary of the training set, and then map the predicted value z from $\boldsymbol{R}$ to a range between 0 and 1. Therefore, the sigmoid function is introduced:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{1}$$

The sigmoid function curve is shown in Figure 2. When $x = 0$, the function value equals 0·5; with the increase of x, the function value will rapidly approach 1; and with the decrease of x, the value of $\sigma(z)$ will rapidly approach 0. If the x-axis scale is large enough, the sigmoid function will be very similar to a step function, but the sigmoid function is easy to deal with in mathematics since it is continuously differentiable. To use this function as a classifier, we need to substitute the predicted value of the linear regression into it (see Equation 2) to obtain a result between 0 and 1, then any result bigger than 0·5 is put into the 1-class, otherwise it is put into the 0-class.

$$y = \frac{1}{1 + e^{-(w^T X + b)}} \tag{2}$$

2.2. *Estimates of regression coefficients.*

If we see y in Equation (2) as the posterior probability of a result in the 1-class occurring, then we get:

$$P(y = 1|X) = \pi(X) = \frac{1}{1 + e^{-(w^T X + b)}} \tag{3}$$

$$P(y = 0|X) = 1 - \pi(X) = \frac{1}{1 + e^{(w^T X + b)}} \tag{4}$$

Here we can use the maximum likelihood method to estimate w and b. For the given training dataset $\{(x_{i1}, x_{i2}, x_{i3}, ..., x_{iN}, y_i)\}_{i=1}^{M}$, the likelihood function of the logistic regression

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Table 1. Comprehensive comparison of common classification algorithms.

	DT	NNs	LR	SVM
Accuracy in general	**	***	****	****
Speed of learning	***	*	***	*
Speed of classification	****	****	****	****
Tolerance to interdependent features	**	***	**	***
Tolerance to irrelevant features	***	*	****	****
Model parameter handling	***	*	*	*
Dealing with danger of over fitting	**	*	***	**
Interpretability	****	*	****	*

(**** represents the best performance and * represents the worst)

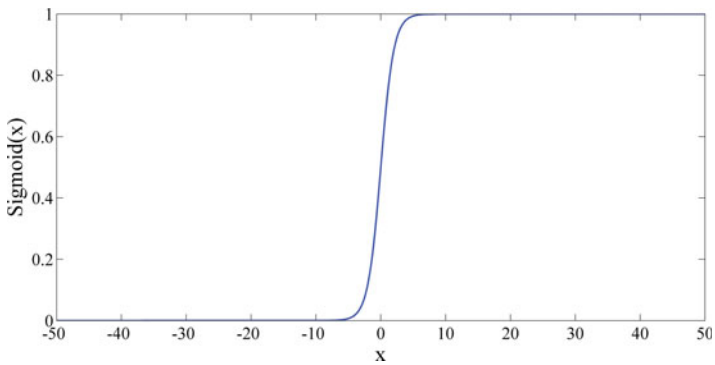

Figure 2. Sigmoid function curve.

model is:

$$L(w,b) = \prod_{i=1}^{M} [\pi(X_i)]^{y_i} [1 - \pi(X_i)]^{1-y_i} \tag{5}$$

Then the log likelihood function is:

$$l(w,b) = \ln L(w,b) = \sum_{i=1}^{M} (y_i \ln[\pi(X_i)] + (1 - y_i) \ln[1 - \pi(X_i)]) \tag{6}$$

Thus, the logistic regression model can be described as an optimisation problem as follows:

$$(w,b)^* = \max_{(w,b)} l(w,b) \tag{7}$$

If there are too many features, over-fitting may occur and the model will lose its generalisation ability. To prevent this phenomenon, a regular item is commonly added to the optimisation target to punish excessive parameters, e.g., the *L1-norm* (sum of absolute value of coefficients) and the *L2-norm* (sum of squares of coefficients). If we add the regular

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

item *L2-norm*, the optimisation target becomes:

$$(w, b)^* = \max_{(w,b)}\{l(w, b) - \lambda \|(w, b)\|_2\} \tag{8}$$

In Equation (8), the parameter λ is the regularisation coefficient. Let $\theta = (w, b)$, $J(\theta) = l(\theta) - \lambda \|\theta\|_2$; the gradient ascent method (or other methods) can be employed to find the maximum value of $J(\theta)$. Then, the optimal estimates of θ will be obtained. The iterative equation of the gradient ascent method is shown in Equation (9). The parameter α is the step length, and $\nabla_\theta J(\theta)$ is the functional derivative. This equation will be executed until a certain termination condition is reached.

$$\theta := \theta + \alpha \cdot \nabla_\theta J(\theta) \tag{9}$$

3. TRAJECTORY PARTITIONING ALGORITHM AND FEATURE EXTRACTION.

Trajectory feature extraction provides a foundation for training the ship type classification model for ship types. In this section, the basic movement patterns of ships are presented first, and then the trajectory partitioning algorithms are designed. Finally, the trajectory feature extraction method is proposed.

3.1. *Definition of basic movement patterns of ships.*

In the existing studies on trajectory data mining, most researchers only partition the trajectory into *stop* and *move* patterns (Li et al., 2008; Yuan et al., 2013; Wang et al, 2014). However, unlike pedestrians, cars or other ground moving objects whose trajectories are restricted by the road network, ships sail on the sea almost unlimitedly, and almost every steering operation has a clear purpose. That is to say, turning is also an important pattern of movement in ship trajectories. On this basis, ship trajectories are divided into three basic patterns in this paper, that is, *anchored-off*, *straight-sailing* and *turning*.

Usually, a vessel anchored at sea has random displacement within a certain range, and there is always error in ship positioning. Hence, the trajectory speed is not necessarily zero when the ship is at rest. The anchored-off pattern of the ship trajectory is defined as follows:

- Definition 1: *Anchored-off pattern*. Assume that a ship trajectory is $T = \{p_1, p_2, p_3, ..., p_n\}$, and p_i is the trajectory point. If there is a sub-trajectory $T' \subseteq T$, the speed of T' is less than δ_v, the duration is greater than δ_t, and the distance between each pair of trajectory points in T' is shorter than δ_d, then the movement pattern of T' is called the *anchored-off* pattern.

For the remaining parts of the trajectory T, their movement pattern is called the *sailing pattern*, but this can be further divided into the *straight-sailing* pattern and the *turning* pattern. In contrast to moving objects on land, the degree of freedom of the ship steering operation is higher, the steering direction may be adjusted back and forth, and the time cost is relatively higher. Thus, the trajectory corresponding to a steering operation is a complex curve. At the same time, the direction may be changed slightly when a ship is sailing on the sea, but this is not discussed in this paper. For this reason, some thresholds are required for filtering. The turning pattern is therefore defined as follows:

- Definition 2: *Turning pattern*. Set the turning threshold as Δ_θ. If the total value of the direction changes before or after a trajectory point p_i within a certain time range

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

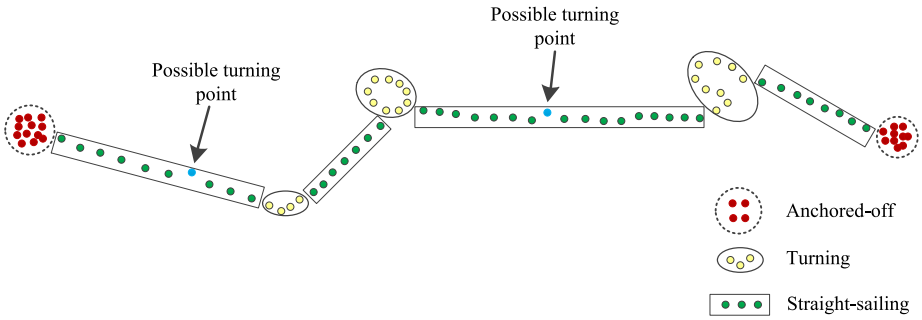

Figure 3. Basic movement patterns of ship trajectory.

Δ_{t1} is greater than Δ_θ, p_i is called the *possible turning point*. If there are several continuous possible turning points and the duration is larger than the threshold Δ_{t2}, the movement pattern of the sub-trajectory consisting of those possible turning points is called the *turning* pattern.

- Definition 3: *Straight-sailing pattern*. The movement pattern of the remaining parts excluding anchored-off and turning pattern trajectories is the so-called straight-sailing pattern. The three defined basic movement patterns of ship trajectory are shown in Figure 3.

3.2. *Trajectory partitioning algorithm.* According to the basic movement patterns defined in Section 3.1, this section studies how to split a trajectory into anchored-off, turning, and straight-sailing sub-trajectories. First, the *SplitTrajs* algorithm is designed to split a complete ship trajectory into anchored-off and sailing sub-trajectories. After that, the *SplitSailingTrajs* algorithm is designed to split the sailing sub-trajectory into turning and straight-sailing segments.

3.2.1. *SplitTrajs algorithm.* The *SplitTrajs* algorithm detects the trajectory points in anchored-off patterns from the given trajectory based on the velocity threshold δ_v, the duration threshold δ_t and the distance threshold δ_d, which are mentioned in Definition 1. Then the trajectory is partitioned into anchored-off and sailing sub-trajectories. The pseudo-code of this algorithm is shown in Figure 4.

The input parameter of *SplitTrajs* is a ship trajectory, and the outputs are a list of anchored-off segments, named *List_TrajStops*, and a list of sailing segments, named *List_TraSails*. In the handling process, the total number of the trajectory points is defined as *length* and the array *sail_state* is initialised to represent the sailing state of each trajectory point. Based on Definition 1, the thresholds δ_v (*SPEED_LIMIT*), δ_t (*TIME_LIMIT*) and δ_d (*DISTANCE_LIMIT*) are set and initialised (lines 1 $\sim$ 3). Next, the velocity of each trajectory point is calculated and recorded in the array *speed*. Then each value in *speed* is compared with *SPEED_LIMIT*. If the *i*th value in this array is less than *SPEED_LIMIT*, then the *i*th value of array *sail_state* is set to 0; otherwise it is set to 1 (lines 4 $\sim$ 10). Third, the segments whose corresponding array *sail_state* continuously has values of 1 are removed. If the interval of one segment is greater than *TIME_LIMIT* and the distance between each pair of points is shorter than *DISTANCE_LIMIT*, this segment is then in an anchored-off pattern and it will be added to the *List_TrajStops* (lines 11 $\sim$ 16). Finally, the remaining segments are added in *List_TraSails* (lines 17 $\sim$ 19).

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

```
ALGORITHM：SplitTrajs
INPUT：Trajectory
OUTPUT: List_TrajStops, List_TrajSails
PROCESS:
1. LET length AS Trajectory.LENGTH
2. LET sail_state[length] AS 0
3. Define SPEED_LIMIT, TIME_LIMIT, DISTANCE_LIMIT
4. Obtain speed[] of each point in Trajectory
5. for i = 1:length
6.    if speed[i] < SPEED_LIMIT
7.       sail_state[i] = 0;
8.    else
9.       sail_state[i] = 1;
10.   end if
11. end for
12. for every continues segment where sail_state[n] to sail_state[m] all equal 0
13.    if time[m] - time[n] > TIME_LIMIT AND min{distance(each two points} < DISTANCE_LIMIT
14.       List_TrajStops.ADD(segment)
15.    end if
16. end for
17. for the rest every continuous segment
18.    List_TrajSails.ADD(segment)
19. end for
RETURN List_TrajStops, List_TrajSails
```

Figure 4. *SplitTrajs* pseudo-code

3.2.2. *SplitSailingTrajs algorithm.* According to Definition 2, the *SplitSailingTrajs* algorithm detects all *possible turning points* first based on the turning threshold Δ_θ and interval threshold Δ_{t1}, and then compares the segment consisting of continuously possible turning points to the duration threshold Δ_{t2} to judge whether it is in a turning pattern. The pseudo-code of this algorithm is shown in Figure 5.

The input parameter of *SplitSailingTrajs* is the sailing sub-trajectory, and the outputs are a list of turning segments, named *List_TrajSails_Turn*, and a list of straight-sailing segments, named *List_TrajSails_Line*. First, the total number of the points in the input segment is set as *length*; array *rotate_state* is initialised to represent the turning state of each trajectory point, and the thresholds Δ_θ (*ANGLE_LIMIT*), Δ_{t1} (*TIME_LIMIT1*) and Δ_{t2} (*TIME_LIMIT2*) are set (lines 1 ∼ 3). Second, the orientation of each trajectory point is calculated (line 4). Third, the total orientation changes in interval Δ_{t1} before and after each point is computed. If any result of p_i is greater than Δ_θ, then it is marked as the *possible turning point* and the *i*th value of *rotate_state* is set to 1(lines 5 ∼ 14). Subsequently, the parts where the corresponding value of *rotate_state* continuously has values of 1 are extracted. If the duration of any part is longer than *TIME_LIMIT2,* then it is in a turning pattern and it will be added to *List_TrajSails_Turn* (lines 16 ∼ 19). Finally, the remaining trajectories are added to *List_TrajSails_Line* (lines 20 ∼ 22).

The threshold parameters in the above algorithms can be determined based on expert experience or experimental results.

3.3. *Trajectory feature extraction.* To study ship classification based on massive trajectory datasets, as many features that can be used to distinguish the ship types are selected as possible. In this section, 17 types of trajectory features are extracted based on the basic movement patterns and the sub-trajectory segments proposed in Sections 3.1 and 3.2. These features can be divided into three categories, that is, global features, straight-sailing features and turning features.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

```
ALGORITHM: SplitSailingTrajs
INPUT: TrajSails
OUTPUT: List_TrajSails_Line, List_TrajSails_Turn
PROCESS:
LET length AS TrajSails.LENGTH
LET rotate_state[length] AS 0
DEFINE ANGLE_LIMIT, TIME_LIMIT1, TIMELIMIT2
Obtain orient[] of each point in TrajSails
for i = 1:length
    Obtain segment where time before time[i] and interval(segment) > TIME_LIMIT1
      if ABS( SUM(all of orient[before]-orient[next] in segment) ) > ANGLE_LIMIT
        rotate_state[i] = 1
      end if
    Obtain segment where time after time[i] and interval(segment) > TIME_LIMIT1
      if ABS( SUM(all of orient[before]-orient[next] in segment) ) > ANGLE_LIMIT
        rotate_state[i] = 1
      end if
end for
for every continues segment where rotate_state[n] to sail_state[m] all equal 1
    if time[m] - time[n] > TIME_LIMIT2
        List_TrajSailsTurn.ADD(segment)
    end if
end for
for the rest every continuous segment
    List_TrajSailsLine.ADD(segment)
end for
RETURN: List_TrajSails_Line, List_TrajSails_Turn
```

Figure 5. *SplitSailingTrajs* pseudo-code.

Table 2. Standard ship course discretisation.

Course range/(°)	Discrete value
337·5 ∼ 22·5	N
22·5 ∼ 67·5	NE
67·5 ∼ 112·5	E
112·5 ∼ 157·5	SE
157·5 ∼ 202·5	S
202·5 ∼ 247·5	SW
247·5 ∼ 292·5	W
292·5 ∼ 337·5	NW

3.3.1. *Global features.* Global features are extracted from all the anchored-off and sailing sub-trajectories, such as the *stop times*, *average duration of stops*, *average speed of sailing*, *total sailing distance* and the *effective distance ratio*. These features can reflect the overall characteristics of ship movement. The *effective distance ratio* is the ratio between the straight-line distance from the start to the end of the ship trajectory and the total sailing distance.

3.3.2. *Straight-sailing features.* The straight-sailing features are extracted from the straight-sailing sub-trajectories, including the *main course*, *main course sailing ratio*, *low speed sailing ratio*, *medium-low speed sailing ratio*, *medium speed sailing ratio* and *high-speed sailing ratio.* To facilitate statistical analysis, it is necessary to discretise the course and speed values (Zhu et al., 2012). Considering how ships actually navigate, the ship course is transformed into eight discrete values, and each of them is separated by 45° (see

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Table 3. Standard ship speed discretisation.

Speed range/(knot)	Discrete value
0 ∼ 4	low
4 ∼ 10	medium-low
10 ∼ 18	medium
18 ∼ 50	high

Table 4. The standard turning angle discretisation.

Turning angle/(°)	Discrete value
0 ∼ 45	small angle
45 ∼ 90	medium angle
90 ∼ 135	large angle
135 ∼ 180	reverse angle

Table 2). Meanwhile, the ship sailing speed is also discretised into four catagories, which are *low*, *medium-low*, *medium* and *high* (Table 3).

According to the discretisation, the features are defined as follows: the *main course* is the discrete value of ship course corresponding to the average direction of the longest straight-sailing segment. The *main course sailing ratio* is the ratio between the total distance of which directions are the same as *main course* among all the straight-sailing segments and the *total sailing distance*. The *low speed sailing ratio* is the ratio between the total time of sailing at low speed among all the straight-sailing segments and the total sailing time. The computation method for *medium-low speed sailing ratio*, *medium speed sailing ratio* and *high speed sailing ratio* is similar to the *low speed sailing ratio.*

3.3.3. *Turning features.* The turning features are extracted from the turning sub-trajectories, including the *maximum turning speed*, *the maximum turning rate*, *small angle turn count*, *medium angle turn count*, *large angle turn count* and *reverse turn count*. For convenience of statistical analysis, the turning angle is transformed into four discrete values as shown in Table 4.

The calculation of turning features are as follows: *Turning speed* is the ratio between the turning distance and the turning time. *Turning rate* is the ratio between the total turning angle and the time of each turning segment. The *small angle turn count* means the total number of turns whose absolute value of direction changed is within the small angle range among all the turning sub-trajectories. The calculations of the *medium angle turn count*, *large angle turn count* and *reverse turn count* are similar to *small angle turn count*.

4. CASE STUDY.

4.1. *Data preparation.*

4.1.1. *Raw data selection.* In this paper, we select real AIS data from the sea area near Shantou, China (Longitude: 116° E ∼ 119° E, Latitude: 21·5° N ∼ 24·5° N) from 1 to 7 October 2015. After data cleaning, data filtering and data transformation, AIS raw data is reconstructed into trajectory data and stored in the database. From the database, 1308 trajectories of fishing boats and cargo ships (654 each) are selected for training and testing the classification model as shown in Figure 6.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

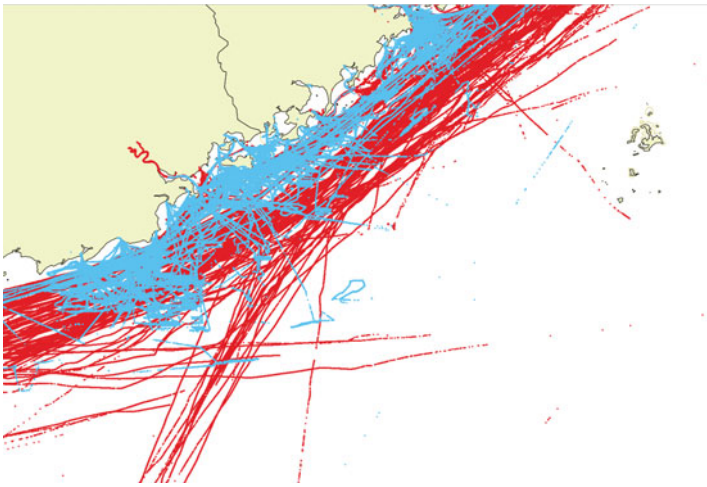

Figure 6. Fishing (blue) and cargo (red) trajectories near Shantou, China.

4.1.2. *Ship sub-trajectory partitioning.* This paper realises the trajectory partitioning algorithm proposed in Section 3.2 by using the Java programming language. By adjusting the threshold parameters, the trajectories can be partitioned into anchored-off, turning and straight-sailing patterns with great accuracy. Some examples are given in Figure 7. On this basis, 10,571 fishing sub-trajectories and 3539 cargo sub-trajectories in total are obtained from the selected trajectories. The results are shown in Table 5.

4.1.3. *Trajectory feature extraction and analysis.* This paper extracts the trajectory features of fishing boats and cargo ships based on the feature extraction methods proposed in Section 3.3 and then builds the feature dataset. The results of trajectory feature analysis are shown in Figure 8.

As shown in Figure 8, the blue line represents the statistical characteristics of fishing boats, and the red line indicates cargo ships. The upper-left sub-graph shows the distribution of the global features by using a box-plot. In this figure, it is clear that cargo ships have *Stop Times* (ST), *Average Duration of Stops* (ADS), *Average Speed of Sailing* (ASS), *total Sailing Distance* (SD) and *Effective Distance Ratio* (EDR) longer or higher than fishing boats. The upper-right sub-graph compares the turning features between the trajectories of fishing boats and cargo ships. There is little difference in the *Maximum Turning Speed* (MTS) and *Maximum Turning Rate* (MTR) between such two types of ships in Figure 8, but the total turn count, including *Small Angle Turn count* (SAT), *Medium Angle Turn count* (MAT), *Large Angle Turn count* (LAT) and *Reverse Angle Turn count* (RAT) in fishing trajectories is significantly higher than for the cargo trajectories. In particular, the LAT and the RAT are more common in fishing boat trajectories, but rarely occur in cargo ship trajectories. The lower-left sub-graph shows the distribution of the main course of the fishing boats and cargo ships in our data. It is found that the *main course* of the cargo ships in this sea area is predominantly in the NE and SW directions, while that of the fishing boats is relatively uniform. Finally, in the lower-right sub-graph we include *Main Course sailing Ratio* (MCR), *Low Speed sailing Ratio* (LSR), *Medium-Low Speed sailing Ratio* (MLSR), *Medium Speed sailing Ratio* (MSR) and *High Speed sailing Ratio* (HSR). It is obvious that the cargo ships have a very high value of MCR, while the sailing speeds are very different

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

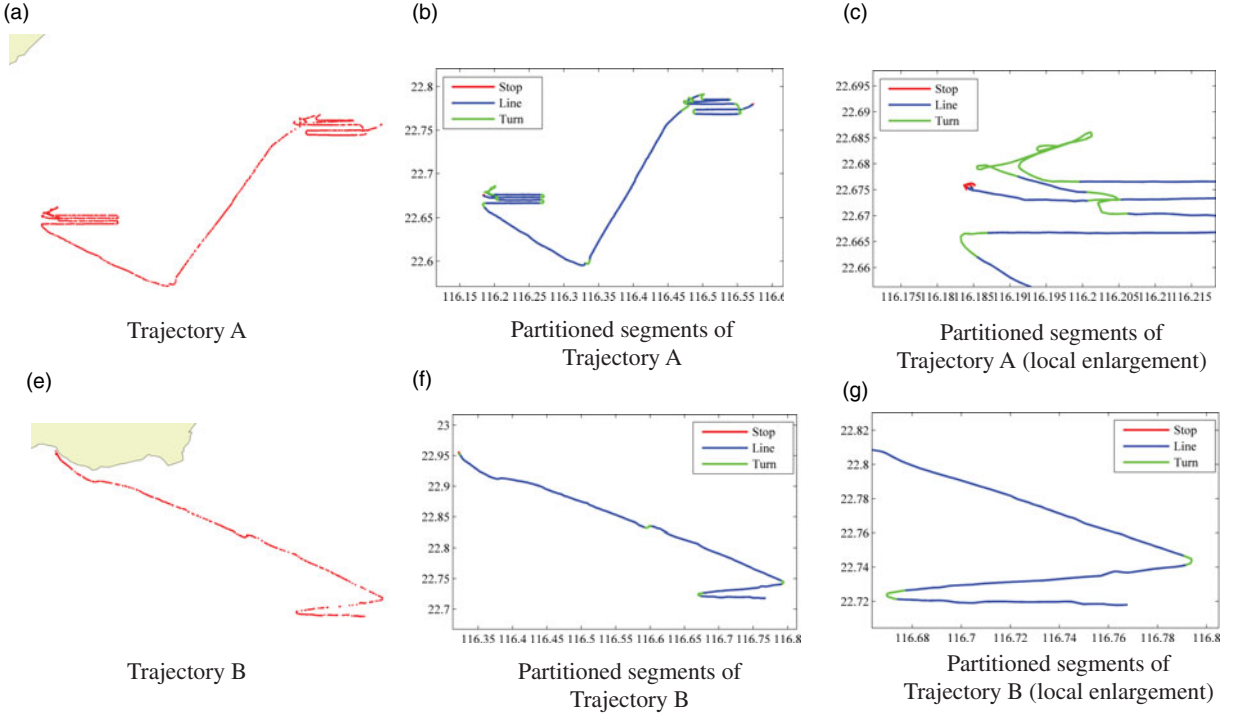

Figure 7. Examples of trajectory partitioning based on real AIS data.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Table 5. The results of ship trajectory partition.

Ship type	Selected trajectories	Anchored-off	Straight-sailing	Turning	Total of sub-trajectories
Fishing	654	435	5077	5059	10571
Cargo	654	521	1587	1386	3494

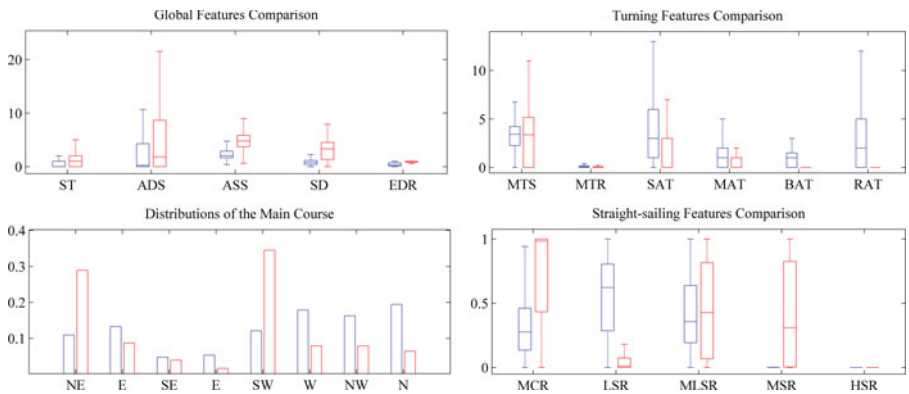

Figure 8. Trajectory feature analysis and comparison of fishing boats (blue) and cargo ships (red).

between these two types of ships. Cargo ships are more likely to be sailing at the MLSR and MSR.

4.2. *Ship classifier training based on logistic regression.* Since this case is a binary classification problem, the logistic regression algorithm is chosen to train the ship trajectory classifier, with the aim of obtaining better predictions and explaining the obtained model. Today, the logistic regression algorithm has matured and has been integrated into a large number of data analysis and mining software packages. In this paper, the classification model is realised by using the Python 2.7 programming language in the *Jupyter Notebook* working environment, and the scientific computing environment is *Anaconda* (Pedregosa et al., 2012). The steps are as follows:

4.2.1. *Feature selection and data conversion.* Based on the feature extraction and analysis methods mentioned above, the two types of ships are not much different from each other in light of such features as maximum turning speed and maximum turning ratio, so the two features are removed to avoid interference. Additionally, features do not have consistent units, so the value of all the features are required to be normalised to $0 \sim 1$ before use. Meanwhile, the classification targets need to be converted into two discrete values, so the class of fishing boat is set to 0 and that of cargo ship to 1.

4.2.2. *Division into training and test data set.* This paper utilises the *hand-out* method to divide the feature data set into two mutually exclusive sets, 75% to the training set and 25% to the test set. To improve the credibility of the model, this paper repeats the experiment 100 times to obtain 100 training sets and their correlated test sets.

4.2.3. *Training and validation of classification model.* In the machine learning integration package *scikit-learn* in the Python environment, the logistic regression model usually requires three parameters, namely, loss function optimisation algorithm *solver* (the method for solving the optimisation target in Equation (8) such as the gradient ascent

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

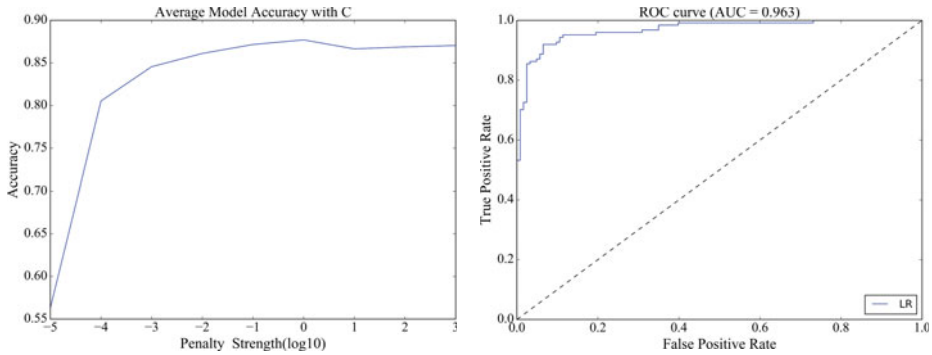

Figure 9. Average model accuracy with the change of *C* (left) and the ROC curve of the optimal classifier (right).

Table 6. Confusion matrix of the classification results.

	Predicted Class	
Actual Class	Cargo ships	Fishing boats
Cargo ships	153	10
Fishing boats	15	148

algorithm in Equation (9)), the penalty function *penalty* (the regular item in Equation (8) such as *L1-norm* and *L2-norm*) and the penalty strength *C* (the inverse of regularisation coefficient λ in Equation (8), with smaller values specifying greater penalties). Due to the small number of data set features, the coordinate descent algorithm (*solver = 'liblinear'*) and *L2* penalisation (*penalty = 'l2'*) are chosen as the loss function optimisation method and the penalty function in this paper, so as to more accurately reflect the contribution of each feature to the final model (Scikit-learn User Guide, 2017). The penalty strength as a parameter needs to be determined experimentally. After a number of experiments (the average model accuracy with the change of *C*, shown in the left graph in Figure 9), the penalty strength is set to 1·0 ($C = 1{\cdot}0$). Under this condition, the classification models are trained 100 times by using the training sets and validated by the corresponding test sets. The average correct rate of all the models is 0·877, the minimum is 0·834 and the maximum is 0·923. The model with the highest test accuracy is selected as the final classification model. The confusion matrix of the classification results of the final model is shown in Table 6.

In the binary classification problem, the Receiver Operation Characteristic (ROC) curve and the Area Under ROC Curve (AUC) are used to evaluate the performance of a classifier. The y-axis of the ROC curve is the True Positive Rate (TPR), which represents the proportion of the positive instances identified by the classifier to all positive instances. The x-axis is the False Positive Rate (FPR), which represents the proportion of negative instances that are divided into the positive class compared to all the negative instances. The ROC curve of a random classifier is the 45° diagonal line, and a perfect classifier has a ROC curve that through the upper left corner (0, 1). In general, the better the classifier, the closer its ROC curve is to the upper left corner. Figure 9 (right) shows the ROC curve of the ship classification model built in this paper. The curve of the classifier is very close to the upper left corner. AUC represents the classifier performance numerically. The AUC of a random classifier is 0·5 and a perfect one is equal to 1·0. By calculation, The AUC value of the

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

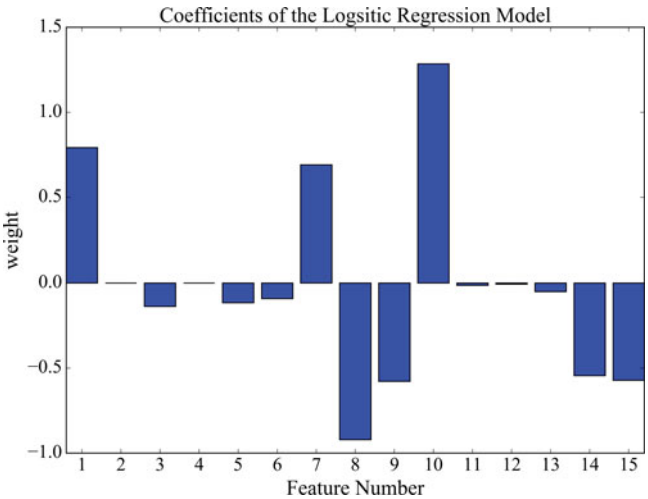

Figure 10. The weight coefficients of logistic regression model (see Table 7 for Feature decode).

final model is 0·963. Therefore, the ship classifier trained by using the presented method has high accuracy and reliability of prediction.

However, some ships are still misclassified. Through analysing raw trajectory data, most misclassified fishing boats sail at a relatively higher speed and the number of large or reverse angle turn count is relatively less. This may be due to fishing boats are not fishing, or they may be just transiting through the selected area for ocean fishing. Comparatively, most misclassified cargo ships are in the berthing state, or the actual sailing distance is very short, since the cargo ships are loading and unloading goods at the port. If the space-time trajectory of the ship is combined with the geographical knowledge and the semantic information is extracted as the classification features, it will be possible to further improve the accuracy of the classification results.

4.3. *Result analysis.* From the bar graph of the weight coefficient of the logistic regression model shown in Figure 10, the contribution of each feature to the ship classifier can be obtained. The positive weight value means this feature affects the cargo class more, while the negative value means the opposite. On the whole, seven out of 15 selected trajectory features are more important, that is, (in the descending order by the absolute value of the weight) *medium speed sailing ratio, low speed sailing ratio, stop times, main course ratio, medium-low speed sailing ratio, reverse angle turn count*, and the *large angle turn count*. The specific weight values of all the trajectory features are shown in Table 7.

5. CONCLUSION.

It is of great significance to identify the type of an unknown ship by its trajectory. This information can be used in many fields, such as camouflage identification, behaviour pattern mining and ship anomaly detection. To solve this problem, this paper makes three main contributions. Firstly, according to the characteristics of ship movement, three kinds of basic ship movement patterns are presented, and the partitioning algorithms for correlated sub-trajectories are designed. Secondly, three categories of ship trajectory features are put forward with their extraction methods. Finally, a ship classification model is built by using a logistic regression algorithm based on ship trajectory features.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Table 7. Weight coefficient of each feature in logistic regression model.

Feature Number	Feature Name	Wight Coefficient
1	Stop times	7·924e−1
2	Average duration of stops	8·402e−6
3	Average speed of sailing	−1·378e−1
4	Total sailing distance	3·423e−4
5	Effective distance ratio	−1·640e−1
6	Main course	−9·115e−2
7	Main course ratio	6·912e−1
8	Low speed sailing ratio	−9·213e−1
9	Medium-low speed sailing ratio	−5·782e−1
10	Medium speed sailing ratio	1·285e+0
11	High speed sailing ratio	−1·338e−2
12	Small angle turn count	−6·745e−3
13	Medium angle turn count	−5·049e−2
14	Large angle turn count	−5·449e−1
15	Reverse angle turn count	−5·723e−1

Through the verification by use of real AIS data, the classification method proposed in this paper can satisfactorily address the classification problem of fishing boats and cargo ships based on their trajectories. However, there is still some work for the future. First of all, more features can be extracted from ship trajectories, such as acceleration and acceleration duration. Generally speaking, the more features, the more accurate the classifier is. Second, the semantic trajectory features can be extracted by combining with the geographic background and ship behaviour patterns and used in the training of future classification models. Third, this paper focuses only on the binary classification problem of ships, but it can be extended to solve the multi-classification problem, so that the research achievement can be better applied in practice. Moreover, the proposed method in this paper only considers off-line applications, but it may be extended for real-time ship classification in two possible ways. One is to dynamically extract the trajectory features of the unknown ships within a defined time window, and then classify and predict the ship type on the basis of the existing ship classification model. The other is to use the incremental learning techniques to modify the ship classifier constantly based on the online data to further improve its timeliness and reliability.

REFERENCES

Aarsæther, K.G. and Moan, T. (2009). Estimating navigation patterns from AIS. *Journal of Navigation*, **62**(4), 587–607.

Boukhechba, M., Bouzouane, A., Bouchard, B., Gouinvallerand, C. and Giroux, S. (2015). Online recognition of people's activities from raw GPS data: semantic trajectory data analysis. *ACM International Conference on Pervasive Technologies Related to Assistive Environments*, 1–8.

De Vries, G.K.D. and Van Someren, M. (2012). Machine learning for vessel trajectories using compression, alignments and domain knowledge. *Expert Systems with Applications*, **39**(18), 13426–13439.

De Vries, G.K.D. and Van Someren, M. (2014). An analysis of alignment and integral based kernels for machine learning from vessel trajectories. *Expert Systems with Applications*, **41**(16), 7596–7607.

Dodge, S., Weibel, R. and Forootan, E. (2009). Revealing the physics of movement: comparing the similarity of movement characteristics of different types of moving objects. *Computers Environment & Urban Systems*, **33**(6), 419–434.

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press

Elwakdy, M., El-Bendary, M. and Eltokhy, M. (2015). A Novel Trajectories Classification Approach for different types of ships using a Polynomial Function and ANFIS. *International Conference on Image Processing, Computer Vision, & Pattern Recognition.*

Feng, Z. and Zhu, Y. (2016). A survey on trajectory data mining: techniques and applications. *IEEE Access*, **4**, 1–1.

Harrington, P. (2012). *Machine Learning in Action*. Manning Publications Co., LTD.

Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learnin*g. Springer.

Kotsiantis, S.B. (2007). Supervised machine learning: a review of classification techniques. *Informatica*, **31**(3), 249–268.

Lee, J.G., Han, J. and Whang, K.Y. (2007). Trajectory clustering: a partition-and-group framework. *ACM SIGMOD International Conference on Management of Data*, pp. 593–604.

Lee, J.G., Han, J., Li, X. and Gonzalez, H. (2008). TraClass: trajectory classification using hierarchical region-based and trajectory-based clustering. *Proceedings of the VLDB Endowment*, **1**(1), 1081–1094.

Li, Q., Zheng, Y., Xie, X., Chen, Y., Liu, W. and Ma, W.Y. (2008). Mining user similarity based on location history. *ACM Sigspatial International Symposium on Advances in Geographic Information Systems, ACM-GIS 2008*, California, USA.

McCauley, D.J., Woods, P., Sullivan, B., Bergman, B. Jablonicky, C., Roan, A., Hirshfield, M., Boerder, K. and Worm, B. (2016). Ending hide and seek at sea. *Science*, **351**(6278), 1148–1150.

Patel, D. (2013). Incorporating duration and region association information in trajectory classification. *Journal of Location Based Services*, **7**(4), 246–271.

Patel, D., Sheng, C., Hsu, W. and Lee, M.L. (2012). Incorporating Duration Information for Trajectory Classification. *IEEE International Conference on Data Engineering*, **41**, 1132–1143.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V. and Thirion, B. (2012). Scikit-learn: machine learning in python. *Journal of Machine Learning Research*, **12**(10), 2825–2830.

Schein, A.I. and Ungar, L.H. (2007). Active learning for logistic regression: an evaluation. *Machine Learning*, **68**(3), 235–265.

Scikit-learn User Guide. (2017). Logistic Regression. http://scikit-learn.org/stable/modules/linear_model.html#logsitic-regression. Accessed 11 May 2017.

Silveira, P.A.M., Teixeira, A.P. and Guedes Soares, C. (2013). Use of AIS Data to Characterize Marine Traffic Patterns and Ship Collision Risk off the Coast of Portugal. *Journal of Navigation*, **66**(6), 879–898.

Wang, Y., Zheng, Y. and Xue, Y. (2014). Travel time estimation of a path using sparse trajectories. *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 25–34.

Yuan, N.J., Zheng, Y., Zhang, L. and Xie, X. (2013). T-finder: a recommender system for finding passengers and vacant taxis. *IEEE Transactions on Knowledge & Data Engineering*, **25**(10), 2390–2403.

Zhen, R., Jin, Y., Hu, Q., Shao, Z. and Nikitakos, N. (2017). Maritime Anomaly Detection within Coastal Waters Based on Vessel Trajectory Clustering and Naïve Bayes Classifier. *Journal of Navigation.* **70**(3), 648–670.

Zheng, Y. (2015). Trajectory data mining: an overview. *ACM Transactions on Intelligent Systems & Technology*, **6**(3), 1–41.

Zhou, Z. (2016). *Machine Learning (in Chinese)*. Tsinghua University press.

Zhu, F., Zhang, Y. and Gao Z. (2012). Research on Ship Behaviors Based on Data Mining (in Chinese). *Navigation of China*, **35**(2), 50–54.

Zhu, Y., Zheng, Y., Zhang, L., Santani, D., Xie, X. and Yang, Q. (2011). Inferring taxi status using GPS trajectories. *Technical Report MSR-TR-2011-144.*

https://doi.org/10.1017/S0373463317000546 Published online by Cambridge University Press